

Prerequisites for Human Development Index (HDI) Prediction Project

1. Programming Knowledge

Python programming, Object-Oriented Programming (OOP), and basic SQL (optional).

2. Machine Learning

Data preprocessing, feature selection, train-test split, regression algorithms, and model evaluation metrics (MAE, MSE, RMSE, R^2).

3. Python Libraries

NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Flask, Joblib.

4. Development Tools

Python 3.x, Visual Studio Code/PyCharm, Jupyter Notebook (optional), Git, GitHub.

5. Dataset

HDI dataset with Country, Year, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per capita, and HDI Value.

6. Software Requirements

Windows/Linux/macOS, Python, VS Code/PyCharm, Git, GitHub, Microsoft Word.

7. Hardware Requirements

Intel Core i3 or above, 4 GB RAM (8 GB recommended), 20 GB free storage.

8. Skills Required

Data cleaning, data visualization, machine learning, Flask, HTML/CSS, GitHub.

9. Project Workflow

Collect dataset → Preprocess → EDA → Train model → Evaluate → Deploy with Flask → Visualize results → Upload to GitHub.